

MATH230: Tutorial Three

Natural Deductions with Classical Logic

Key ideas

- Write natural deductions using RAA.
- Prove LEM and DNE.
- Use LEM and DNE as derived rules of inference to avoid RAA directly.
- Witness the oddities of classical theorems.

Relevant Topic: Propositional Logic

Relevant reading: L $\exists\forall$ N Chapter 5

Hand in exercises: 2a, 2b, 3a, 3b, 3e

Due Friday @ 5pm to the submission box on Learn.

Discussion Questions

1. $\neg(A \wedge B) \vdash \neg A \vee \neg B$

Tutorial Exercises

1. **NOTE!** Make sure you have finished all of the minimal and intuitionistic natural deductions before doing this tutorial. It is more important that you understand those.

2. **LEM and DNE.** Prove each of the following fundamental theorems of classical logic making explicit use of the RAA mode of reasoning:

(a) $\vdash A \vee \neg A$ [Challenge!]

(b) $\neg\neg A \vdash A$

3. **Classical derivations.** Provide natural deduction proofs of the following. All rules *may* be required. Rather than making explicit use of RAA, it can be easier to appeal to LEM or DNE as *derived rules of inference*.

(a) $\neg(A \wedge B) \vdash \neg A \vee \neg B$

(b) $A \rightarrow B \vdash \neg A \vee B$

(c) $\vdash (A \rightarrow B) \vee (B \rightarrow C)$

(d) $\vdash A \rightarrow B \vee \neg B$

(e) $\vdash (\neg A \rightarrow A) \rightarrow A$

(f) $(A \rightarrow B) \vdash (A \rightarrow D) \vee (C \rightarrow B)$

(g) $\neg(A \rightarrow B) \vdash A \wedge \neg B$

(h) $\vdash ((A \rightarrow B) \rightarrow A) \rightarrow A$

[Challenge!]

(i) $\vdash ((A \rightarrow B) \rightarrow B) \rightarrow A$

4. In class we discussed how classical logic can be obtained from intuitionistic logic by adding the following rule of inference *reductio ad absurdum*: If $\frac{\Sigma}{\perp} \mathcal{D}$ is a deduction of $\perp$ from Σ , then

$$\frac{\frac{\frac{\overline{\neg\alpha}}{\Sigma} \mathcal{D}}{\perp}}{\alpha} \text{ RAA}$$

is a derivation of α from the assumptions $\Sigma \setminus \{\neg\alpha\}$.

In this question we will explore this extension of logics in more detail. We will see that there are different methods for obtaining classical logic from minimal/intuitionistic logic. For the purposes of this question we introduce two more rules of inference:

Double Negation Elimination

$$\frac{\frac{\neg\neg\alpha \rightarrow \alpha}{\neg\neg\alpha} \quad \frac{\Sigma}{\mathcal{D}}}{\alpha} \text{ DNE}$$

Law of Excluded Middle

$$\frac{\frac{\Sigma_1}{\mathcal{D}_1} \quad \frac{\Sigma_2}{\mathcal{D}_2} \quad \frac{\alpha \vee \neg\alpha \quad \alpha \rightarrow \gamma \quad \neg\alpha \rightarrow \gamma}{\gamma}}{\gamma} \text{ LEM}$$

Question: Suppose you are given a proof witnessing the sequent $\Sigma, \neg\alpha \vdash \perp$.

- Using only minimal logic + DNE extend this proof to a proof witnessing the sequent $\Sigma \vdash \alpha$.
- Using only intuitionistic logic + LEM extend this proof to a proof witnessing the sequent $\Sigma \vdash \alpha$.

Use part (a) to argue that $\text{RAA} = \text{DNE}$ in the presence of minimal logic. Where as part (b) shows $\text{RAA} = \text{LEM}$ in the presence of intuitionistic logic. Finally, argue that all three are therefore equivalent modes of reasoning in the presence of intuitionistic logic: $\text{RAA} = \text{DNE} = \text{LEM}$.